Literacy Trend Intelligence Engine

Project Overview

The Literacy Trend Intelligence Engine is an interactive data analytics platform developed using Python and Streamlit.

The platform helps users analyze literacy trends across countries and forecast future literacy rates using machine learning.

Objectives

- Analyze literacy data
- Compare countries
- Identify growth trends
- Forecast future literacy rates
- Generate downloadable reports

Technologies Used

- Python
- Streamlit
- Pandas
- Plotly
- Scikit-Learn
- NumPy

Dataset

Source:

UNESCO Literacy Dataset

Columns:

- Country

- Year
- Literacy Rate

Features

Dashboard

Displays:

- Total Countries
- Total Records
- Earliest Year
- Latest Year

Trend Analysis

Visualizes literacy growth over time.

Country Comparison

Compares literacy rates between multiple countries.

AI Forecasting

Uses Linear Regression to predict future literacy rates.

Reports

Download country-wise CSV reports.

Machine Learning Model

Algorithm:

Linear Regression

Purpose:

Predict literacy trends for next 5 years.

Project Structure

```
Literacy_Trend_Intelligence_Engine
|
├── app.py
├── literacy_data.csv
├── requirements.txt
├── README.md
|
├── models
|   └── predictor.py
|
├── pages
|    ├── Dashboard
|    ├── Trend Analysis
|    ├── Country Comparison
|    ├── AI Forecasting
|   └── Reports
```

Results

The platform successfully provides:

- Data visualization
- Comparative analysis
- Predictive analytics
- Downloadable reports

Future Enhancements

- Advanced forecasting models

- Interactive world map
- PDF report generation
- Real-time UNESCO data integration

Conclusion

The Literacy Trend Intelligence Engine demonstrates how data analytics and machine learning can be used to monitor and forecast global literacy improvements.